

Can We Trust LLMs for Mental Health Screening?

Consistency, ASR Robustness, and Evidence Faithfulness

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Abstract

LLMs can estimate Hospital Anxiety and Depression Scale (HADS) scores from speech in a zero-shot manner, but clinical deployment requires reliability across three dimensions: intra-model consistency, ASR robustness, and evidence faithfulness. We evaluate three LLMs (Phi-4, Gemma-2-9B, and Llama-3.1-8B) on 111 English-speaking participants using ground-truth transcripts and three Whisper ASR variants (Large, Medium, Small), with three independent runs per model-condition pair. We find that (i) Phi-4 and Gemma-2-9B achieve excellent intra-model consistency ($ICC > 0.89$) with minimal degradation under ASR; (ii) Llama-3.1-8B shows ASR-fragile consistency, with ICC dropping from 0.82 to 0.36 at 10% WER; (iii) predictive validity is largely preserved under ASR for robust models; and (iv) keyword groundedness exceeds 93% for Phi-4 and Gemma-2-9B but falls to 77–81% for Llama-3.1-8B. Inter-model keyword agreement is far lower than score-level agreement, revealing a score–evidence dissociation with implications for clinical interpretability.

Index Terms: Mental health, LLM, HADS, depression, anxiety, ASR, consistency, reliability, keywords

1. Introduction

Mental health disorders such as anxiety and depression impact millions worldwide each year [1]. Although early detection is critical for effective intervention [2], traditional screening relies on clinical interviews [3] or the Hospital Anxiety and Depression Scale (HADS) [4], approaches that are resource-intensive, subjective, and difficult to administer at scale [5]. While HADS scores are obtainable via questionnaire, self-report relies on active participation, introspective access, and willingness to disclose, assumptions that fail for populations affected by stigma, cognitive decline, or language barriers.

Speech-based estimation offers a scalable alternative by sidestepping these barriers, enabling passive screening embedded in routine interactions and longitudinal monitoring free of questionnaire fatigue. More fundamentally, as the field moves toward integrating language models into clinical workflows, reliable *screening* is a practical prerequisite for downstream applications such as treatment monitoring. Accurate HADS estimation therefore serves as a validation gate for establishing broader clinical utility.

A rich body of work has established that language carries reliable markers of psychological states: depression manifests through negative emotion words, first-person pronouns, and reduced lexical diversity [6, 7], while anxiety is associated with hedging, uncertainty expressions, and hesitation markers [8].

These insights have driven text-based screening from classical machine learning on social media [9, 10] and clinical interviews through deep learning approaches [11, 12]. In the speech domain, acoustic features have been successfully used for depression and emotion recognition [13, 14, 15], and multimodal systems combining audio and text further improve detection [16].

Large language models (LLMs), whose zero-shot capabilities were established by Brown et al. [17], have extended these approaches to estimating psychological states without task-specific training. LLM-based systems have shown promise for depression detection from social media [18, 10], clinical dialogues [19, 20], and personal narratives [21], with systematic reviews confirming their growing role [22, 23]. Zero-shot mental health classification can approach supervised baselines [24], and interpretability-focused prompting strategies further improve clinical utility [25]. It was also demonstrated that diverse LLMs can estimate HADS scores from speech transcripts with moderate validity (Spearman $\rho_s = 0.38$ – 0.58) [26, 27], while Roy et al. [28] showed zero-shot diagnostic performance comparable to fine-tuned models. However, these studies evaluate only predictive accuracy, and methodological concerns persist [29]: *reliability*, *robustness*, and *interpretability* remain largely unexamined, yet are indispensable for clinical adoption.

Three specific gaps motivate this work. First, LLM inference is inherently stochastic, yet the reproducibility of clinical predictions across repeated runs has received little attention. Second, while LLMs can cite evidence for their predictions, chain-of-thought explanations may be unfaithful to the model’s actual decision process [30], and hallucinated clinical evidence could mislead clinicians [31], making evidence groundedness verification imperative [32]. Third, practical deployment requires automatic speech recognition (ASR), yet modern ASR systems like Whisper [33] systematically delete disfluencies and hesitation markers in clinical speech [34], precisely the features LLMs may rely on for psychological assessment, and the interaction between ASR errors and downstream clinical predictions remains unexplored.

We address these gaps with four research questions, whose systematic investigation forms this paper’s core contributions:

1. Are LLM predictions stable across repeated runs, and does ASR affect this stability?
2. Does ASR degrade predictive validity (correlation with HADS ground truth)?
3. Are LLM-cited keywords grounded in the transcript, and is keyword evidence stable across runs and models?
4. Are there model-specific vulnerability patterns that inform deployment decisions?

Having introduced the dataset, HADS instrument, and

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prompt design in Section 2, we present the statistical analysis framework underpinning our evaluation in Section 3. Section 4 describes the experimental setup. Section 5 presents results as well as discussion and Section 6 concludes the paper.

2. Data, HADS, and Prompt Design

2.1. The PsyVoiD corpus

We use the PsyVoiD corpus [35], collected in Scotland (UK) during the COVID-19 lockdown to investigate the relationship between spontaneous English speech and psychological traits. The dataset comprises 111 participants (70 female, 41 male), aged 21–86 (median 62), 34 of whom (31%) report a prior clinical history of depression (not necessarily current). Participants recorded one-to-two-minute spontaneous speech samples via personal devices in home environments, describing their daily lives during lockdown, yielding transcripts with considerable length variability (mean 152 words, $SD = 77$). Each participant completed the Hospital Anxiety and Depression Scale (HADS) questionnaire [4] on the same day as the recording, providing paired ground-truth anxiety and depression scores. The study was approved by the NHS Health Research Authority (REC reference: 20/EM/0146); all participants provided informed consent.

2.2. The HADS instrument

HADS [4] is a widely validated self-report instrument [36, 37], yielding two subscale scores: Anxiety (HADS-A) and Depression (HADS-D), each ranging from 0 to 21, with established clinical thresholds (0–7 normal, 8–10 borderline, 11–21 clinical). In PsyVoiD, HADS-A has a mean of 6.7 ($SD = 4.9$) and HADS-D has a mean of 5.0 ($SD = 4.5$). While all three severity tiers are represented, the distribution is skewed toward the normal range, with relatively few participants in the clinical tier ($HADS \geq 11$).

2.3. Prompt design

To maximise reliability and interpretability, we designed a structured prompt informed by clinical psychology and linguistics expertise [26]. Figure 1 shows the complete prompt. It embodies four design principles:

- *role specification*: assigning the LLM a dual expert persona (clinical psychologist and linguist) to activate domain-relevant knowledge [38, 22];
- *step-by-step decomposition*: structuring the analysis into four sequential steps (psychological feature extraction, linguistic analysis, score prediction, keyword justification) to enforce chain-of-thought reasoning [17, 38] before scoring;
- *keyword justification*: requiring the model to cite specific transcript keywords, producing an auditable evidence trail and constraining predictions to be grounded in the input [32];
- *structured JSON output*: requested via the prompt (not enforced through constrained decoding) to minimise free-text variability and facilitate automated parsing.

3. Statistical Analysis Framework

In this section we describe the measures used to evaluate the three reliability dimensions. All statistical tests are non-parametric, reflecting the ordinal nature of HADS scores.

You are a clinical psychologist and linguist, analyzing a spontaneous speech transcript recorded during the Covid-19 lockdown. Your task is to estimate the speaker's psychological state using linguistic, emotional, and psychological analysis. Specifically, you will predict the Hospital Anxiety and Depression Scale (HADS) scores.

Analysis Steps

- 1. Psychological and Emotional Features**
 - Detect anxiety-related cues (e.g., excessive worry, nervousness).
 - Detect depression-related cues (e.g., lack of motivation, hopelessness).
- 2. Linguistic and Behavioural Features**
 - Identify hesitation markers (e.g., erm, uh, pauses) and their frequency.
 - Detect negative self-statements (e.g., I do not feel good, I cannot cope).
 - Assess certainty level (confident, unsure, detached).
- 3. Psychological Score Predictions**
 - Predict HADS Anxiety score (0-21 scale, higher indicates greater anxiety)
 - Predict HADS Depression score (0-21 scale, higher indicates greater depression)
- 4. Justification Using Keywords**
 - Provide keywords from the transcript that influenced each prediction.

Transcript: {transcript}

Figure 1: Zero-shot prompt used for HADS score estimation, integrating role specification, step-by-step decomposition, score prediction, and keyword justification. Adapted from [26].

3.1. Intra-model consistency

To assess whether repeated LLM inference yields stable predictions, we use the Friedman test to detect systematic inter-run differences, and ICC(3,1) (two-way mixed, single measures, consistency [39]) to quantify inter-run agreement. ICC thresholds follow Koo and Li [40]: <0.50 poor, 0.50 – 0.75 moderate, 0.75 – 0.90 good, >0.90 excellent.

3.2. Predictive validity and ASR robustness

To evaluate predictive validity and quantify the impact of ASR errors on LLM-based HADS estimation, we use the following complementary measures:

- *Spearman's rank correlation* (ρ_s) measures the monotonic association between LLM predictions, computed as the mean of three inference runs, and HADS ground truth. It captures rank-order validity without assuming linearity or normality.
- *Wilcoxon signed-rank test* assesses systematic distributional differences between predicted and ground-truth HADS scores. A significant result ($p < 0.05$) indicates consistent over- or under-estimation by the model.
- *ASR robustness metrics* directly quantify the effect of transcription errors by comparing ground-truth-based and ASR-based predictions for each subject using mean absolute error (MAE) and paired ρ_s .

3.3. Keyword evidence analysis

To evaluate the faithfulness and stability of the transcript evidence cited by LLMs in support of their predictions, we analyse keyword behaviour along three complementary dimensions:

- *Groundedness* measures whether cited keywords are present in the source transcript and is assessed using exact case-insensitive substring matching, supplemented by fuzzy matching based on the Levenshtein partial ratio (≥ 80) implemented via `rapidfuzz`.
- *Intra-model keyword stability* is quantified using Jaccard similarity between keyword sets across three independent runs for each subject, capturing whether a model consistently cites the same evidence when re-run under identical conditions.
- *Inter-model keyword agreement* is measured using Jaccard similarity across different models for each subject, quanti-

fying the extent to which distinct LLMs ground their predictions in the same transcript evidence.

4. Experimental Setup

4.1. LLM configurations

We evaluate three open-weight instruction-tuned LLMs, selected to span distinct model families, training pipelines, and parameter scales: Phi-4 (14.7B, Microsoft [41]), Gemma-2-9B (9B, Google [42]), and Llama-3.1-8B-Instruct (8B, Meta [43]). Following [26, 27], each model receives the zero-shot prompt (Section 2.3) containing the transcript and produces HADS-A and HADS-D scores, and supporting keywords. For each model-transcript combination, we perform three independent inference runs with default sampling (temperature ≈ 0.7) to quantify intra-model variability. We deliberately retain stochastic sampling rather than setting temperature to 0, because characterising intra-model variability under realistic deployment conditions is itself a central research question of this study.

4.2. Transcript conditions

Each model is evaluated under four transcript conditions: (1) GT: ground-truth human transcription (WER = 0%); (2) W-Large: Whisper Large-v3 (WER = 8.6%); (3) W-Medium: Whisper Medium-en (WER = 9.3%); (4) W-Small: Whisper Small-en (WER = 10.1%). The three Whisper variants were chosen to span a range of model capacities and WER levels within a single ASR family, isolating the effect of transcription quality from architectural differences. Note that these WER values reflect Whisper’s performance on spontaneous elderly speech, which is higher than the WER typically reported on read-speech benchmarks. The Whisper error profile on PsyVoID is deletion-heavy (4.1–4.6% deletion rate) [26], preferentially removing hesitation markers and disfluencies common in elderly spontaneous speech [34]. Each Whisper variant was run once to produce a single deterministic transcript per speaker; the three inference runs vary only the LLM sampling, not the ASR output. This yields 3 models $\times$ 4 conditions $\times$ 3 runs = 36 experimental configurations.

5. Experimental Results and Discussion

5.1. Consistency across runs

Table 1 reports ICC(3,1) and Friedman p -values across all model-condition-subscale combinations. Phi-4 and Gemma-2-9B show excellent consistency with minimal degradation across ASR conditions. Phi-4’s ICC ranges from 0.890 to 0.925 across all conditions and subscales ($\Delta_{\max} = 0.035$). Gemma-2-9B achieves even higher ICC (0.947–0.964), with a maximum cross-condition variation of only 0.017. Neither model shows significant Friedman effects in any condition (all $p > 0.05$), confirming that no run systematically deviates.

Llama-3.1-8B emerges as the critical outlier. Under GT transcripts, its intra-model consistency is borderline acceptable (HADS-A: 0.753; HADS-D: 0.818). Under ASR conditions, however, consistency degrades sharply. At W-Small (10.1% WER), ICC drops to 0.358 for HADS-A and 0.381 for HADS-D, falling below the “poor” reliability threshold of 0.50 [40]. The Friedman test is also significant in this condition ($p = 0.04$ for HADS-A; $p = 0.03$ for HADS-D), indicating systematic differences across inference runs rather than random sampling noise. These results suggest that ASR errors interact with

Table 1: *Intra-model consistency: ICC(3,1) and Friedman p -value per model, transcript condition, and HADS subscale (A = Anxiety, D = Depression). Bold: ICC ≥ 0.85 ; † : $p < 0.05$.*

Model	HADS	ICC / Friedman p			
		GT	W-Large	W-Med.	W-Small
Phi-4	A	.898 / .43	.900 / .51	.890 / .78	.897 / .44
	D	.925 / .17	.895 / .66	.914 / .85	.903 / .32
Gemma-9B	A	.964 / .44	.955 / .58	.954 / .62	.955 / .12
	D	.947 / .96	.948 / .46	.949 / .89	.953 / .52
Llama-8B	A	.753 / .10	.709 / .65	.728 / .21	.358 / .04 †
	D	.818 / .46	.741 / .70	.697 / .69	.381 / .03 †

Llama-3.1-8B’s internal representations in a manner that amplifies stochastic variability, whereas Phi-4 and Gemma-2-9B remain robust to such perturbations. The tight coupling between score-level consistency and keyword stability (Gemma-9B > Phi-4 > Llama-8B; see Section 5.3) further reinforces the conclusion that reliable predictions cannot be achieved in the absence of stable and consistent evidence generation.

5.2. Robustness to ASR quality

Table 2 and Figure 2 show Spearman correlations between model predictions and HADS ground truth across transcript conditions. Phi-4 exhibits an unexpected pattern for HADS-A: validity *increases* from $\rho_s = 0.469$ (GT) to 0.503 (W-Small). Whisper’s deletion-heavy error profile removes disfluencies (“erm,” false starts) that Phi-4 over-interprets as anxiety features, as confirmed by keyword frequency analysis where “erm” dominates Phi-4’s HADS-A keyword list (Figure 3). Since PsyVoID participants skew older (median age 62) and disfluency patterns in elderly spontaneous speech correlate with arousal rather than anxiety [34], their removal inadvertently corrects a systematic bias, with ASR acting as a beneficial denoising filter for this model. For HADS-D, validity decreases mildly (0.564 $\rightarrow$ 0.511, $\Delta = 0.053$). This paradox is model-specific: Gemma-2-9B shows a gentle monotonic decline for both subscales (HADS-A: 0.476 $\rightarrow$ 0.409; HADS-D: 0.476 $\rightarrow$ 0.379), and Llama-3.1-8B experiences the steepest HADS-A decline (0.451 $\rightarrow$ 0.338, $\Delta = 0.113$), paralleling its ICC collapse. Depression validity is generally more resilient, consistent with depression markers being more lexically robust than prosodically-mediated anxiety cues [13, 8].

Table 3 provides the direct subject-level comparison between GT-based and ASR-based predictions. Phi-4 and Gemma-2-9B produce GT-vs-ASR predictions with $\rho_s = 0.75$ –0.87 and MAE of 0.49–1.00 points, suggesting that for these models, ASR-induced shifts are small relative to the prediction scale. Llama-3.1-8B shows substantially poorer agreement (MAE up to 2.06 points, ρ_s as low as 0.57), confirming that ASR perturbations interact destructively with its inference.

Pairwise inter-LLM agreement (ρ_s between models’ mean predictions) ranges from 0.62 to 0.87 (all $p < 10^{-4}$). Phi-4–Gemma-9B achieves the strongest agreement ($\rho_s = 0.87/0.84$ for HADS-A/D under GT), while Llama-8B pairs are weakest under ASR ($\rho_s = 0.62$ –0.65 at W-Small), mirroring its intra-model instability. These results highlight that model selection dominates ASR condition as a source of variation: the reliability gap between Phi-4/Gemma-9B and Llama-8B far exceeds the difference between any two Whisper variants, and even smaller

Table 2: Predictive validity: Spearman ρ_s between LLM predictions (mean of 3 runs) and HADS ground truth. All $p < 0.001$.

Model	HADS	GT	W-Large	W-Med.	W-Small
Phi-4	A	0.469	0.430	0.496	0.503
	D	0.564	0.496	0.496	0.511
Gemma-9B	A	0.476	0.451	0.443	0.409
	D	0.476	0.449	0.398	0.379
Llama-8B	A	0.451	0.429	0.346	0.338
	D	0.475	0.484	0.451	0.422

Table 3: GT-vs-ASR prediction agreement: MAE, Spearman ρ_s , and % of predictions within 1 point. All use mean-of-3-runs.

Model	ASR	MAE		Spearman ρ_s		% ≤ 1
		A	D	A	D	
Phi-4	W-L	0.87	0.87	0.810	0.873	73.1
	W-M	0.74	0.77	0.753	0.820	78.8
	W-S	0.81	0.75	0.780	0.843	80.7
Gemma-9B	W-L	0.86	0.49	0.867	0.828	79.2
	W-M	1.00	0.51	0.848	0.871	79.7
	W-S	0.93	0.52	0.845	0.871	78.8
Llama-8B	W-L	1.78	1.46	0.669	0.678	44.3
	W-M	1.97	1.45	0.588	0.618	37.3
	W-S	2.06	1.72	0.573	0.596	45.8

ASR models suffice (ρ_s difference between W-Large and W-Small ≤ 0.07 for reliable LLMs).

5.3. Keyword groundedness

Beyond scores, each LLM outputs transcript keywords to justify predictions. Table 4 summarises three evidence dimensions. Phi-4 and Gemma-2-9B achieve >93% groundedness, meaning fewer than 1 in 14 keywords are fabricated. Llama-3.1-8B is substantially less grounded (77–81%), frequently generating abstract clinical labels (“lack of motivation,” “social withdrawal”) rather than verbatim transcript evidence.

Intra-model keyword stability follows the same rank ordering as ICC (Gemma-9B > Phi-4 > Llama-8B), confirming that score and reasoning consistency are tightly coupled. A consistent anxiety–depression asymmetry emerges: keyword stability is universally lower for HADS-D (intra Jaccard: 0.21–0.62) than for HADS-A (0.41–0.67), and inter-model keyword agreement for depression (0.07–0.17) is roughly half that for anxiety (0.22–0.25). This aligns with depression manifesting through latent lexical features [8, 7, 16] that are harder to capture with consistent keywords than the more overt markers of anxiety.

The most striking finding is the score–evidence dissociation: inter-model keyword Jaccard ranges from 0.07 to 0.25, yet score-level ρ_s reaches 0.87, a gap of 0.5–0.7 showing that models converge on *what* to predict far more than on *why* they predict it (Figure 3). This implies that presenting a single model’s keywords as “the reason for this score” overstates reasoning certainty; only keywords corroborated across models or runs should be treated as robust evidence [30, 27].

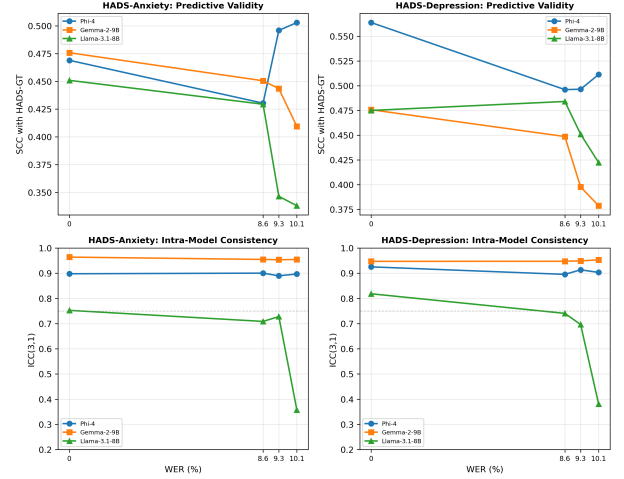


Figure 2: Top: Predictive validity (Spearman ρ_s with HADS ground truth) vs WER. Bottom: Intra-model consistency (ICC) vs WER. Note the sharp ICC decline for Llama-3.1-8B at higher WER, contrasting the stability of Phi-4 and Gemma-2-9B.

Table 4: Keyword analysis (GT transcripts): groundedness, intra-model Jaccard (3 runs), and mean inter-model Jaccard (A = Anxiety, D = Depression).

Model	Grounded. (%)		Intra Jaccard		Inter Jaccard	
	A	D	A	D	A	D
Phi-4	93.0	94.3	0.48	0.38	0.25	0.15
Gemma-9B	94.3	94.6	0.67	0.62	0.23	0.10
Llama-8B	77.4	80.8	0.41	0.21	0.23	0.12

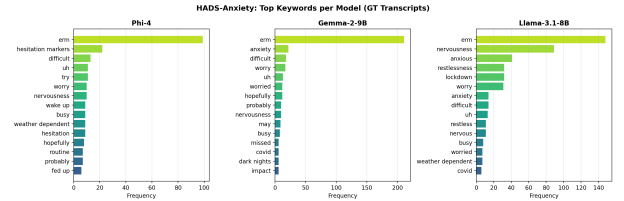


Figure 3: Top 15 keywords per model for HADS-Anxiety (GT transcripts). Note the dominance of “erm” across models and divergent keyword vocabularies underlying similar predictions.

6. Conclusions and Future Work

We presented the first joint analysis of intra-model consistency, ASR robustness, and keyword evidence faithfulness for LLM-based mental health screening. Phi-4 and Gemma-2-9B demonstrate excellent consistency (ICC>0.89) and stable predictive validity ($\rho_s = 0.38$ –0.56) across ASR conditions, whereas Llama-3.1-8B exhibits severe consistency degradation under ASR. Keyword analysis shows that reliable score predictions are accompanied by more grounded (>93%) and stable evidence, but inter-model keyword agreement (Jaccard ≤ 0.25) remains far lower than score-level agreement (ρ_s up to 0.87), revealing a pronounced score–evidence dissociation. These results establish model-specific ASR tolerance and evidence faithfulness as essential criteria for deployment and support fully automated Speech $\rightarrow$ ASR $\rightarrow$ LLM screening pipelines for mod-

els that satisfy all three reliability dimensions.

From a clinical perspective, the results indicate that reliability is best improved through model selection rather than aggressive ASR optimisation, and that three-run averaging is needed to mitigate stochastic variability. Evidence presented to clinicians should be automatically checked for groundedness and corroborated across runs or models, with particular caution for depression-related keywords. We note that PsyVoID was collected during a pandemic lockdown with a sex and age imbalance (63% female, median age 62) and relatively few participants in the clinical severity range, which may limit generalisability. Future work should evaluate broader ASR error regimes, additional ASR architectures, and more diverse populations, investigate whether stability converges beyond three runs or requires more, and examine whether multimodal approaches combining acoustic and textual features can improve robustness and interpretability for anxiety detection.

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